

Inflation-Linked Bond Pricer

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model: "inflation_linked"
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Government linkers — TIPS, OATi, BTPi, UK Gilts

Description

Plain-vanilla government-style inflation-linked bond pricer. Covers US TIPS, French OATi/OAT€i, Italian BTP-i, UK Gilts, and similar instruments where both coupons and redemption are scaled by the ratio of the current CPI to the base CPI at issuance. The redemption is optionally floored at par (standard for TIPS). Also computes breakeven inflation if a nominal yield is provided.

Pricing Methodology

Forward index ratios are projected from the current known ratio using a flat annual inflation assumption:

$$\text{IndexRatio}(d) = (\text{current_cpi} / \text{base_cpi}) \times (1 + \text{annual_inflation})^t$$

Each real coupon and the real redemption are multiplied by the index ratio at the payment date, then discounted with the benchmark curve + z-spread. Breakeven inflation is the rate that equates the inflation-adjusted NPV to the price implied by the nominal yield.

Key Formulas

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IndexRatio(d) = (current_cpi/base_cpi) * (1+pi)^yearFraction(eval,d)
coupon(t) = real_coupon * IndexRatio(t) * par * accrual
redemption = max(par, par * IndexRatio(T)) [if redemption_floor=true]
NPV = Σ coupon(t) · DF*(t) + redemption · DF*(T)
breakeven: solve pi* s.t. NPV(pi*) = par / (1+nominal_yield)^T
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Required Fields

Field	Type	Description
instrument_id	string	ISIN or internal identifier
evaluation_date	string	Pricing date
issue_date	string	Issue date
maturity_date	string	Maturity date
real_coupon_rate	float	Annual real coupon rate
base_cpi	float	CPI at issuance (index ratio denominator)
current_cpi	float	Current / latest CPI level
annual_inflation_rate	float	Forward flat inflation assumption
coupon_frequency	string	Annual Semiannual Quarterly Monthly
accrual_day_count	string	ACT/ACT (ICMA) typical for linkers
calendar	string	TARGET UnitedStates

Optional Fields

Field	Type	Description
par	float	Face value (default 100)
credit_spread_bp	float	Z-spread in basis points (default 0)
redemption_floor	bool	Floor accreted redemption at par (default True)
nominal_yield	float	If provided, breakeven inflation is computed
settlement_days	int	Days to settlement (default 2)

Key Outputs

- npv / dirty_price, clean_price, accrued
- index_ratio_at_maturity — projected final CPI ratio

- $\text{inflation_accreted_redemption} = \text{par} \times \text{IndexRatio}(T)$
- $\text{breakeven_inflation}$ — if nominal_yield was provided
- ytm — continuously compounded real yield
- cashflows — per-period real coupons $\times$ index ratio